

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



Department of Computer Science and Engineering

Course No: CSE 2104

Course Title: Data Structures Lab

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CSE2104 Offline-6

Write a program to support multi-digit numbers in the conversion of an infix expression to postfix notation and the evaluation of the resulting postfix expression. For example, your code should convert this infix expression to its appropriate postfix notation: $(12 + 34) * 56 - 78 / 9$. Then evaluate the result of this expression from the postfix notation.

Solution

```
#include <bits/stdc++.h>
using namespace std;

int calc(int a, int b, char op)
{
    if(op == '/') return a/b;
    if(op == '*') return a*b;
    if(op == '+') return a+b;
    if(op == '-') return a-b;
    return 0;
}

int prec(char op)
{
    if (op == '*' || op == '/') return 2;
    if (op == '+' || op == '-') return 1;
    return 0;
}

string infixToPostfix(string s)
```

```

{
    stack<char> st;
    string post = "";

    for(int i = 0; i < s.size(); i++)
    {
        char c = s[i];

        if( c == ' ') continue;
        if( c >= '0' && c <= '9')
        {
            while(i < s.size() && s[i] >= '0' && s[i] <= '9')
            {
                post += s[i];
                i++;
            }
            post += " ";
            i--;
        }
        else if(c == '(')
        {
            st.push('(');
        }
        else if (c == ')') {
            while (!st.empty() && st.top() != '(') {
                post += st.top();
                post += " ";
                st.pop();
            }

```

```

        st.pop();
    }
    else {
        while (!st.empty() && prec(st.top()) >= prec(c)) {
            post += st.top();
            post += " ";
            st.pop();
        }
        st.push(c);
    }

}

while (!st.empty()) {
    post += st.top();
    post += " ";
    st.pop();
}

return post;
}

```

```

int evaluatePostfix(string post) {
    stack<int> st;

    for (int i = 0; i < post.length(); i++) {
        char c = post[i];

        if (c == ' ') continue;
        if (c >= '0' && c <= '9') {

```

```

    int num = 0;
    while (i < post.length() && post[i] >= '0' && post[i] <= '9') {
        num = num * 10 + (post[i] - '0');
        i++;
    }
    st.push(num);
    i--;
}
else {
    int b = st.top(); st.pop();
    int a = st.top(); st.pop();
    st.push(calc(a, b, c));
}
}

return st.top();
}

int main()
{
    string infix = "(12 + 34) * 56 - 78 / 9";

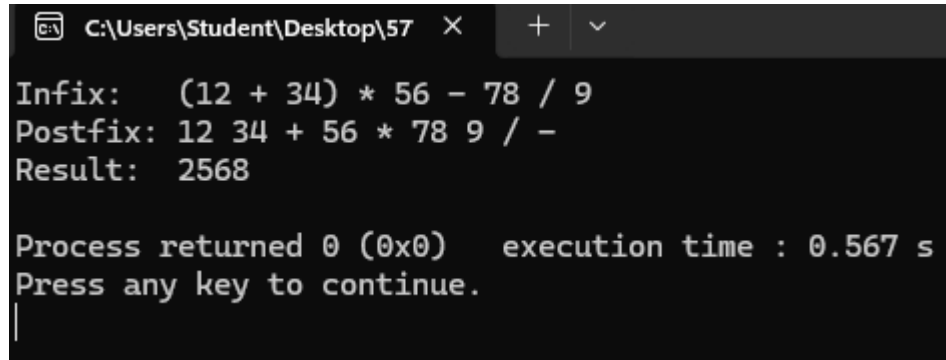
    string postfix = infixToPostfix(infix);
    int result = evaluatePostfix(postfix);

    cout << "Infix: " << infix << endl;
    cout << "Postfix: " << postfix << endl;
    cout << "Result: " << result << endl;
}

```

```
return 0;  
}
```

Output



```
C:\Users\Student\Desktop\57  X  +  v  
Infix:  (12 + 34) * 56 - 78 / 9  
Postfix: 12 34 + 56 * 78 9 / -  
Result: 2568  
  
Process returned 0 (0x0)  execution time : 0.567 s  
Press any key to continue.  
|
```